Seat No.:_____

AHU ID No._____ Name_____ Major_____ Grade_____

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Stony Brook Institute at Anhui University

2023 Fall Semester

Probability Theory and Mathematical Statistics Midterm Exam

(Closed-book Exam Duration: 120 minutes)

Section	I	II	III	IV	Total score
Score					
Grader					

I. Blank Fillings (5 items, 2 points each)

Score	

1. Consider two random events A and B with $P(A)=0.3$, $P(B)=0.5$, $P(A|B)=P(B|\bar{A})$, then $P(AB)=$_____.

2. A box contains 100 balls, of which r are red. Suppose that the balls are drawn from the box one by one at random without replacement. Then the probability that the 50*th* ball drawn will be red is _____.

3. Suppose that the random variable $X \sim G(0.5)$, now conduct three independent observations of X, then the probability that at least 2 observations of X is larger than 2 is _____.

4. Suppose that the random variable X follows binomial distribution with $n=10000$, $p=0.0002$, then the **approximate** value of $P(X \geqslant 3)$ is_____.

5. Suppose that (X, Y) obeys the uniform distribution in region $D=\{(x, y): 0<x<2, 0<y<x\}$, then the joint probability density function $f(x, y)$ is _____.

II. Choices (5 items, 2 points each)

Score	

6. Suppose that random event A, B are independent, $0<P(A)<1$, $0<P(B)<1$, then the wrong option is ( ).

A. Event A and B are exclusive. B. $P(AB)=P(A)P(B)$.

C. $P(A|\bar{B})=P(A)$. D. $P(B|A)=P(B)$.

7. Suppose that the probability that draw a red ball from the box is p. If someone picks a red ball, he will have a lucky chance to buy ice-cream at a very good price. Suppose also that each person has ten times to draw balls from that box with replacement, then the probability that someone happens to get his/her third lucky chance on the 10th draw is ( ).

A. $C_{10}^{3}p^{3}(1-p)^{7}$ B. $C_{9}^{2}p^{3}(1-p)^{7}$

C. $C_{9}^{3}p^{2}(1-p)^{7}$ D. $C_{10}^{2}p^{3}(1-p)^{7}$

8. Suppose that David has 5 keys and only 1 key can open the door. He will try one of these keys to open the door at will, and remove each key after trying it once, then the probability that he can open the door in the third time is ( ).

A. 1/5 B. 4/125 C. 20/125 D. 16/125

9. Suppose that the probability mass function of discrete random variable X is $P(X=k)=b*0.2^{k}, (k=1,2,3,\ldots)$, then the value of unknown constant b is ( ).

A. 1 B. 4 C. 6 D. 3

10. Suppose that r.v.$(X,Y)\sim N(\mu_1,\mu_2,\sigma_1^2,\sigma_2^2,\rho)$, then the number of correct conclusions in the following conclusions about (X,Y) is ( ).

① $X\sim N(\mu_1,\sigma_1^2)$ ② $Y\sim N(\mu_2,\sigma_2^2)$ ③ X, Y are independent

A. 0 B. 1 C. 2 D. 3

III. Calculations (6 items, 12 points each)

Score	

11. In a certain city, 30 percent of the people are Conservatives, 40 percent are Liberals, and 30 percent are Independents. Records show that in a particular election. 60 percent of the Conservatives voted, 80 percent of the Liberals voted, and 50 percent of the Independents voted.

a. If a person in the city is selected at random, what is the probability that she **did not** vote in the last election?

b. If a person in the city is selected at random, and it is learned that she **did not** vote in the last election, what is the probability that she/he is a Liberal?

12. Suppose that the p.d.f. of a random variable X is as follows:

$$\boldsymbol{f}(\boldsymbol{x}) = \begin{cases} x^2, & 0 < x < 1 \\ x, & 1 \leq x < \mathrm{c} \\ 0 & \text{otherwise.} \end{cases}$$

a. Find the value of the constant c and the c.d.f $F(x)$.

b. Find the value of $P(1/2 < X < 3/2)$.

c. Find the value of constant t such that $P(X<t) = 2/3$.

13. Suppose the r.v. X and Y are independent, the joint probability mass function of random variable (X, Y) as the following table:

X \ Y	**0**	**1**	**2**	**P(X=x_i)**
1		1/8		1/4
2	1/8			
P(Y=y_j)	1/6			

a. Filling the missing probability in the table.
b. Find the marginal distribution of r.v. X and Y, respectively.
c. Find P(X+2Y>2).

14. Suppose the joint probability density function of random variable (X,Y) is

$$f(x,y) = \begin{cases} kx, & 0 < x < y < 1 \\ 0, & \text{else} \end{cases}$$

a. Find the marginal p.d.f of X and Y. Are X, Y independent?

b. Find P($2X+Y<1$).

15. Suppose that the joint probability density function of random variable (X, Y) is:

$$f(x,y) = \begin{cases} \frac{3}{4}(x^2 + 3y^2), & 0 < \boldsymbol{x} < 1, 0 < y < 1 \\ 0, & else \end{cases}$$

a. Find the marginal c.d.f. of $F_x(x)$.

b. Find the conditional p.d.f of $f(y|x)$ and $\mathrm{P}(Y < 1/3 | X = 1/3)$.

16. Suppose that the height of junior boys（unit: cm）in Hefei is r.v. X, and its obeys Normal distribution with $X \sim N(150, 10^2)$. Now suppose also that a junior boy in Hefei is randomly selected. Here some values you may need: $\Phi(0.5)=0.6915$, $\Phi(1.0)=0.8413$, $\Phi(1.5)=0.9332$, $\Phi(1.28)=0.8997$, $\Phi(1.29)=0.9015$.

a. Find the probability that his height is between (135, 160).

b. Find the median height of the junior boys in Hefei.

c. What is a boy's height if his height exceeds 90% Hefei junior boys?

IV. Analytical Judgement (8 points)

Score	

17. Suppose that r.v. X obeys an Exponential distribution with parameter $\lambda=2$.
    a. Find $P(X<8|X>3)$.
    b. Find the probability the equation $4t^2 + 4tX + X + 2 = 0$ has real roots.